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Wu et al.

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(54) **BATTERY, POWER CONSUMPTION
DEVICE, METHOD FOR PRODUCING
BATTERY**

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H01M 50/289; H01M 50/291

See application file for complete search history.

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Primary Examiner — Scott J. Chmielecki

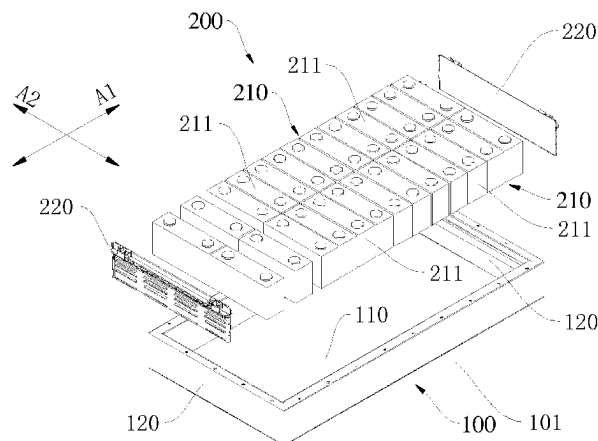
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(57) **ABSTRACT**

The present application provides a battery, a power consumption device, and a method for producing a battery. The battery includes a box body and a battery module arranged in the box body, the battery module includes a battery cell arrangement structure and an end plate, the battery cell arrangement structure includes a plurality of battery cells stacked along a first direction, the end plate is arranged between an inner wall of the box body and the battery cell arrangement structure, and the end plate is fixedly connected to the battery cell arrangement structure, where the end plate includes a first elastic support portion, and the first elastic support portion is capable of abutting against the inner wall of the box body and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands, to provide expansion space for the battery cell arrangement structure.

17 Claims, 8 Drawing Sheets

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(52) **U.S. Cl.**

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2220/20 (2013.01)

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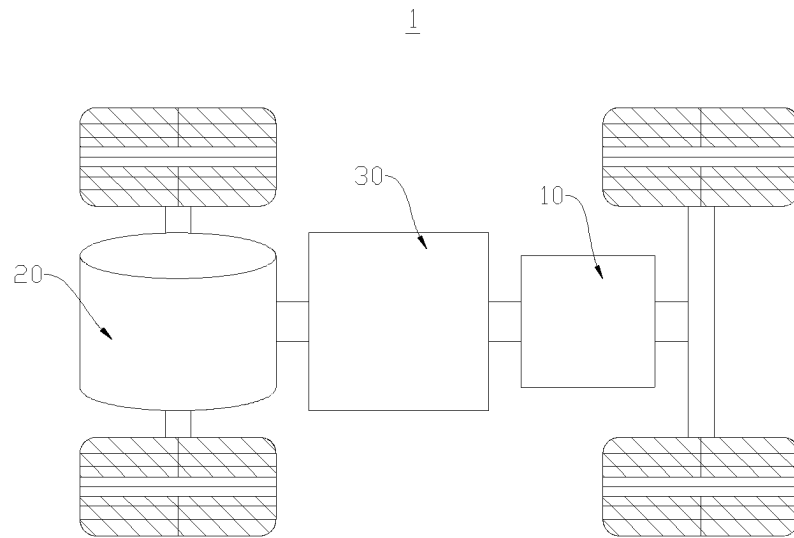


FIG. 1

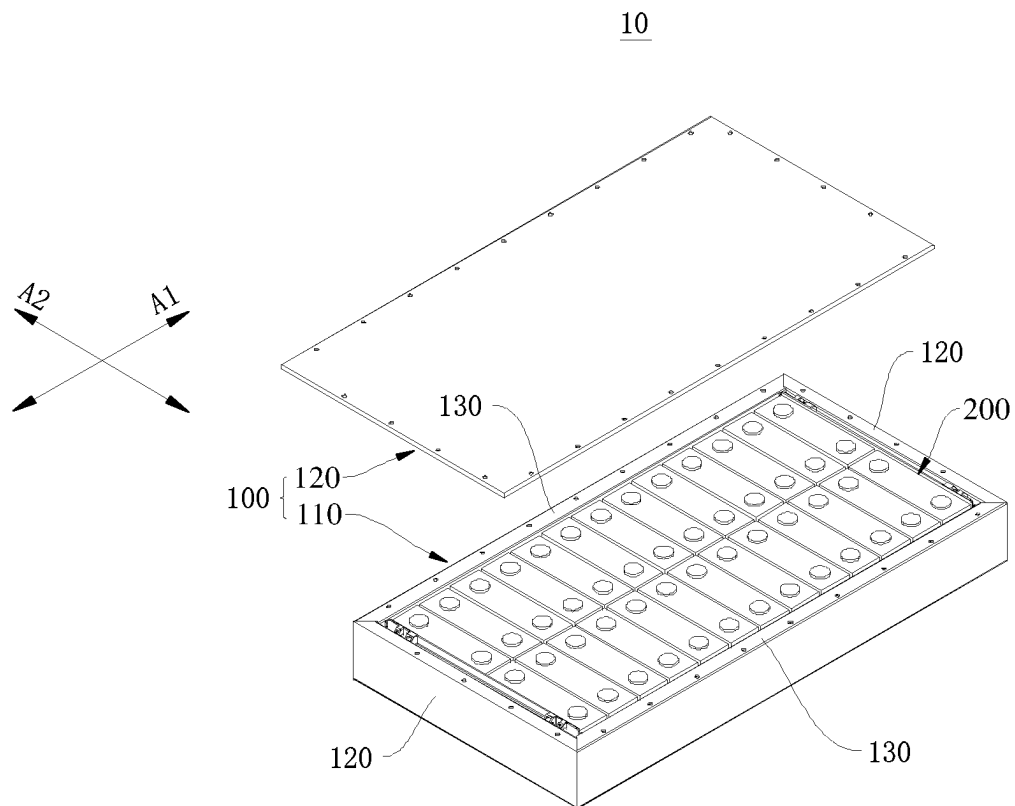


FIG. 2

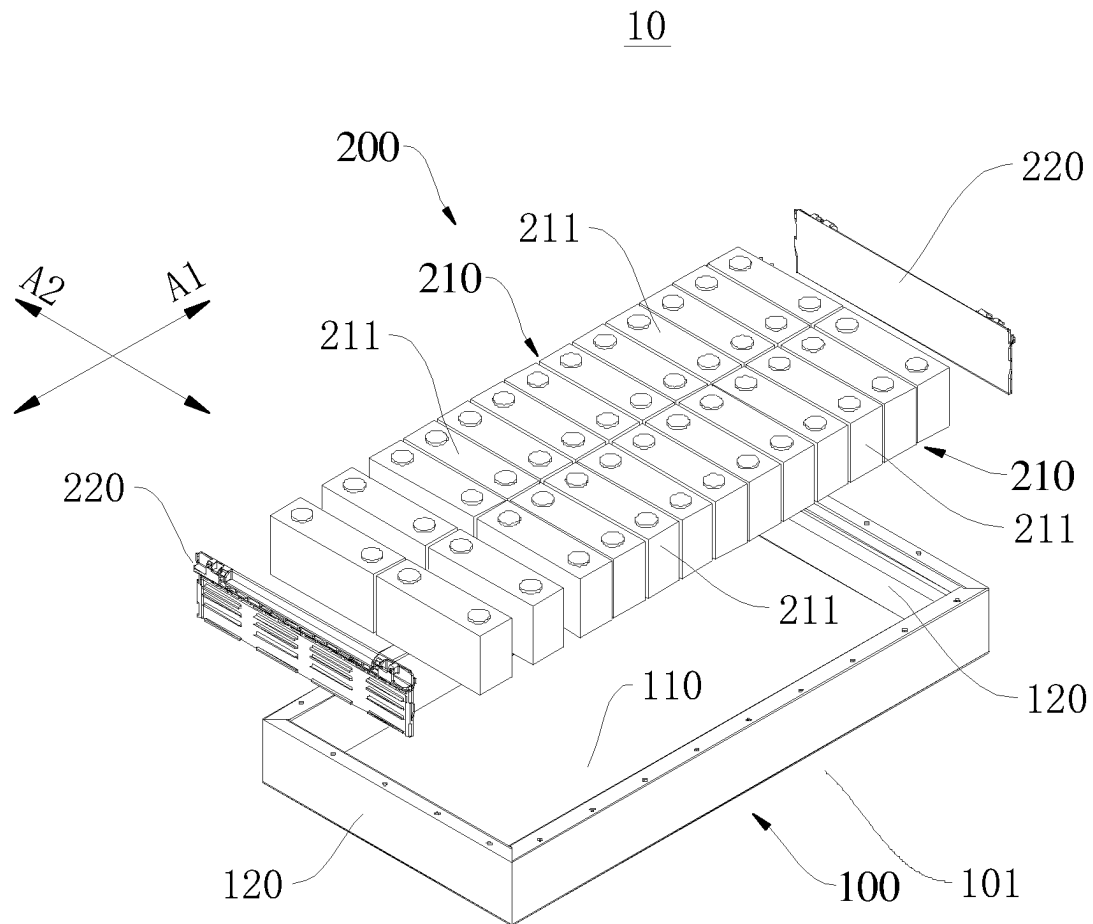


FIG. 3

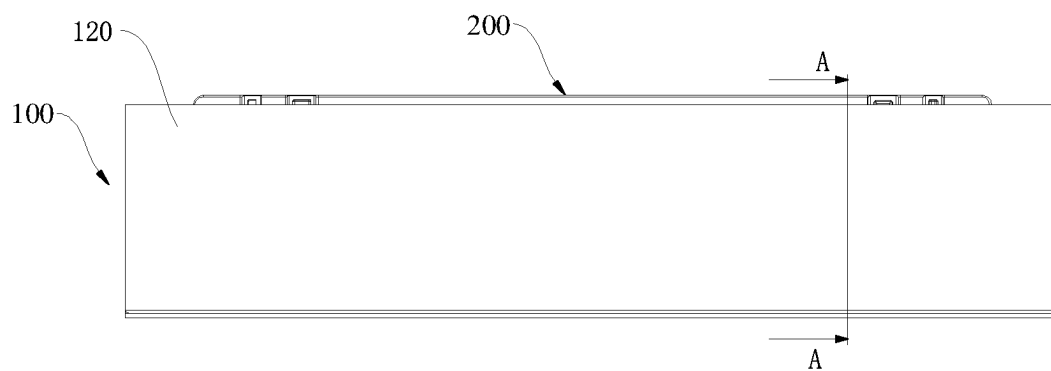


FIG. 4

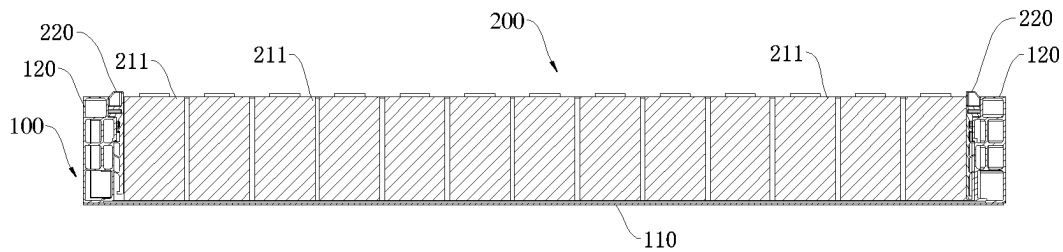


FIG. 5

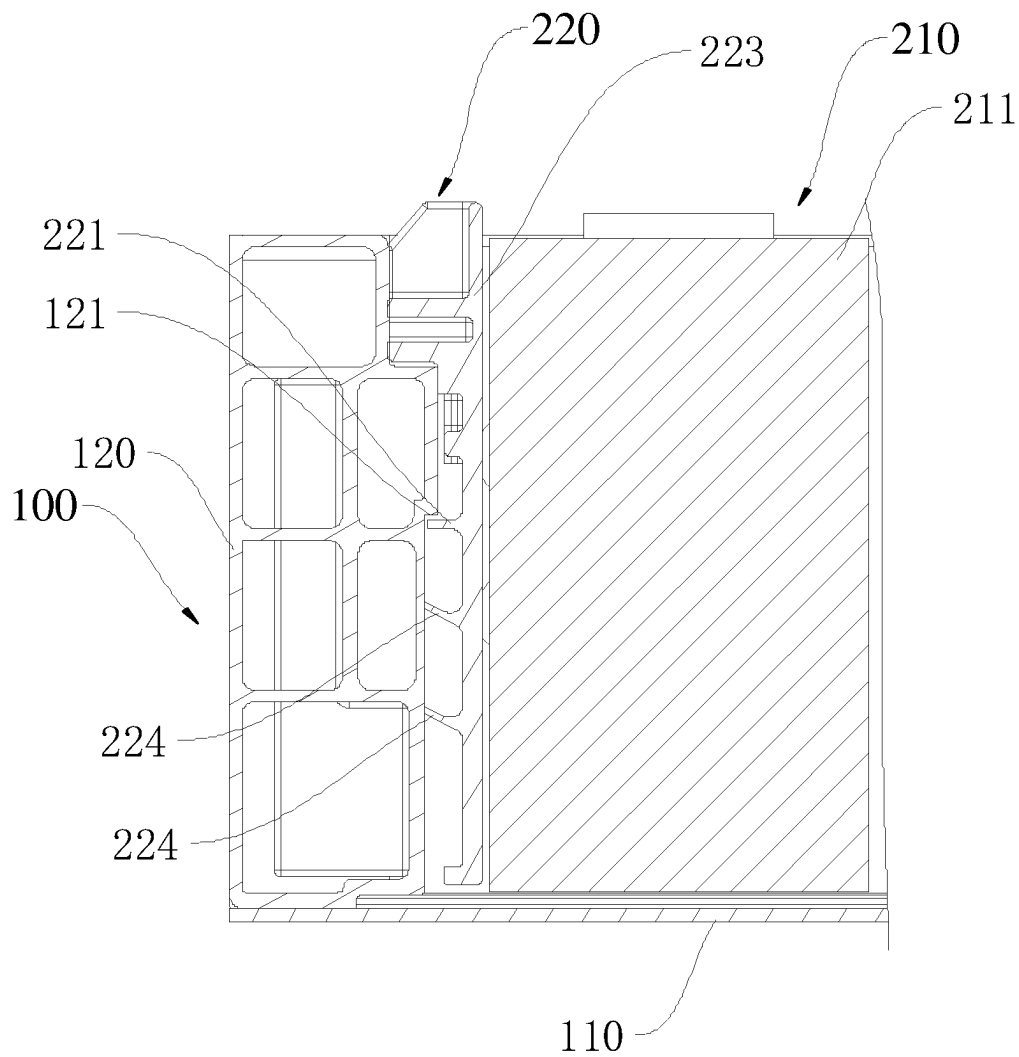


FIG. 6

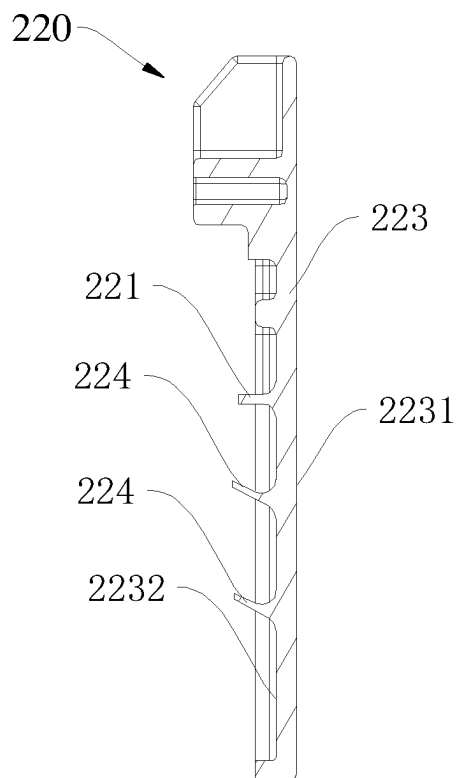


FIG. 7

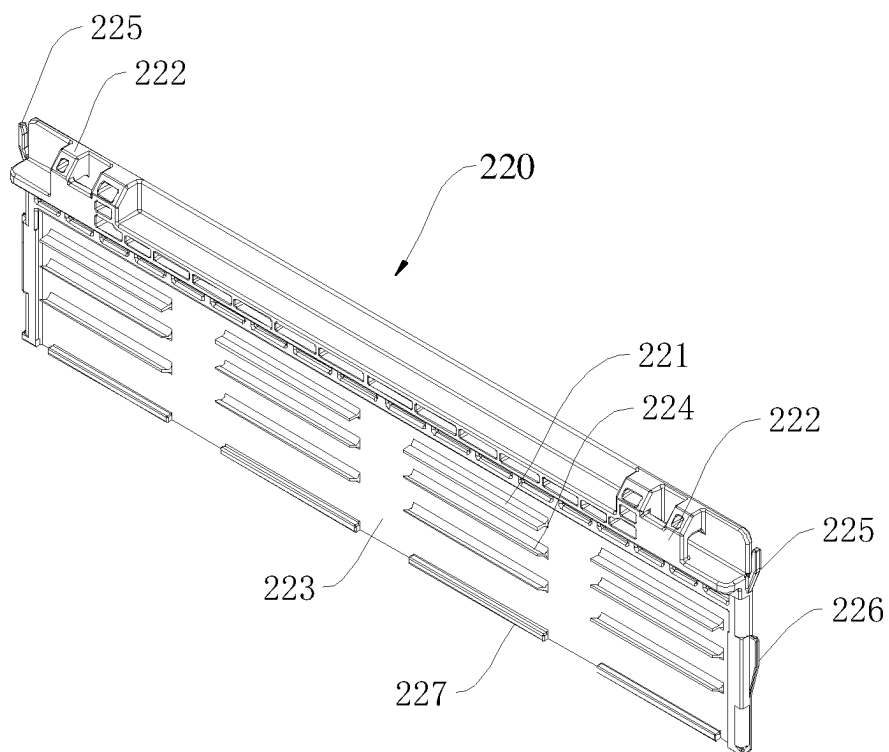


FIG. 8

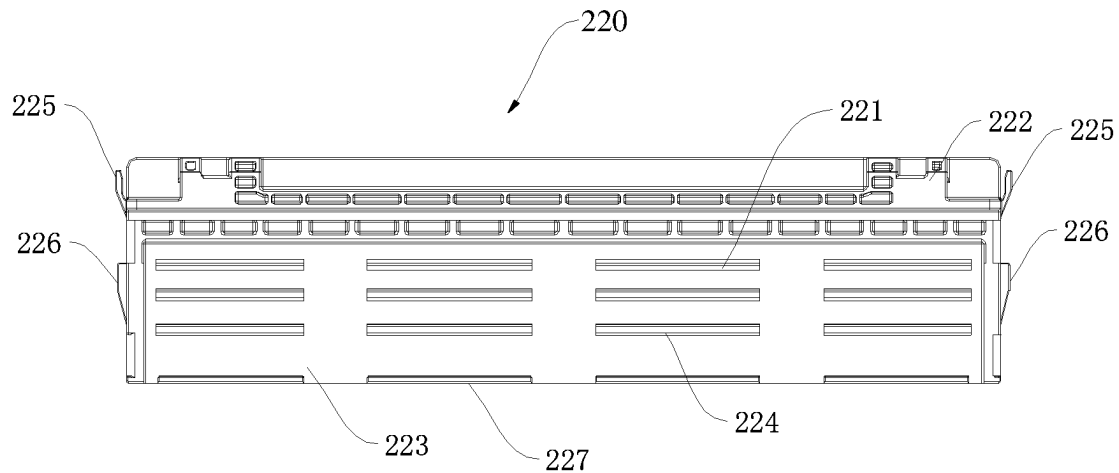


FIG. 9

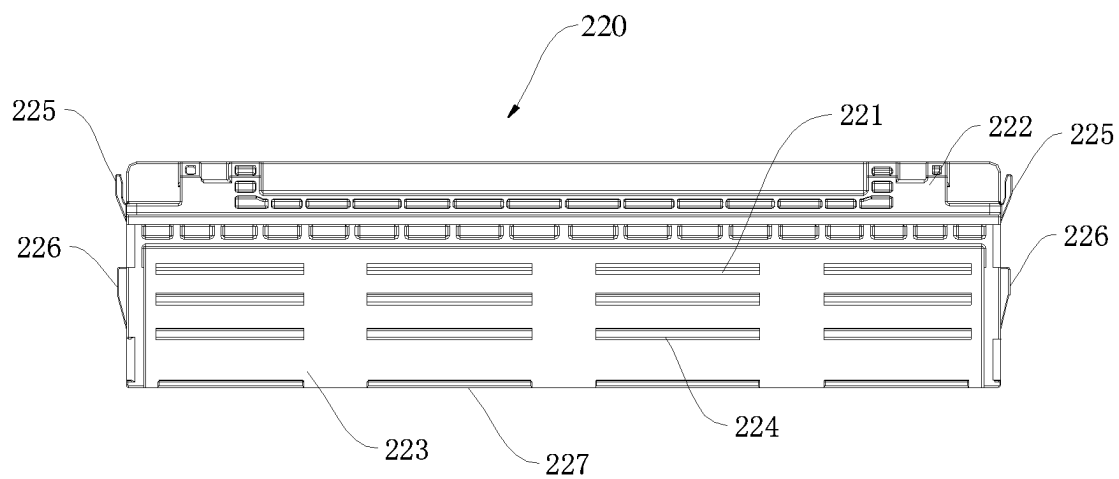


FIG. 10

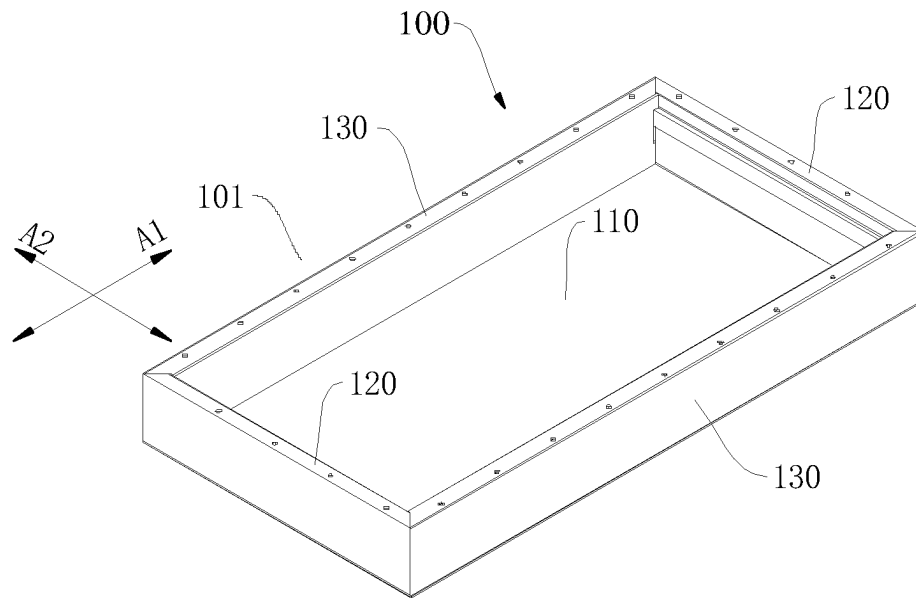


FIG. 11

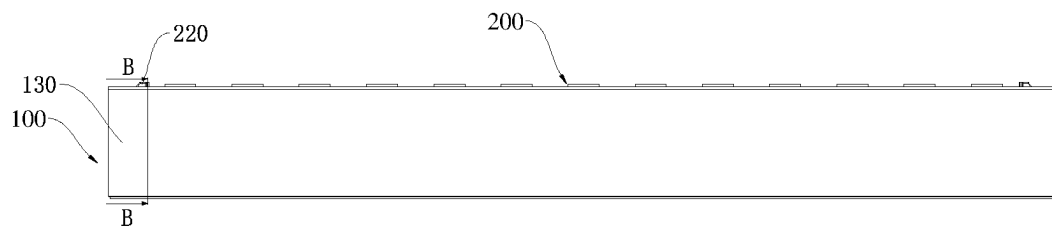


FIG. 12

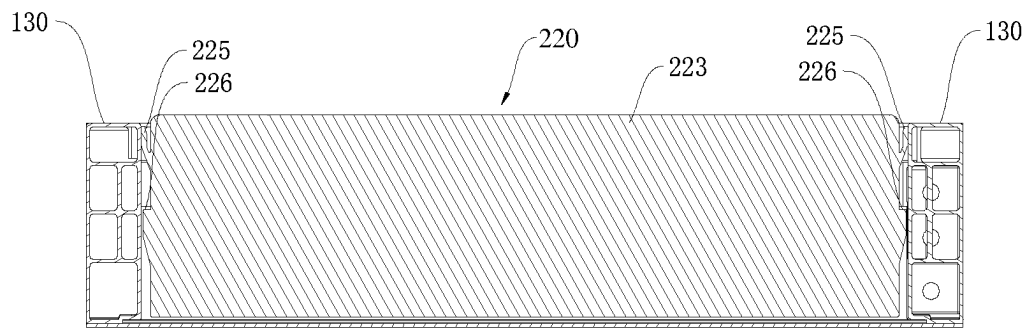


FIG. 13

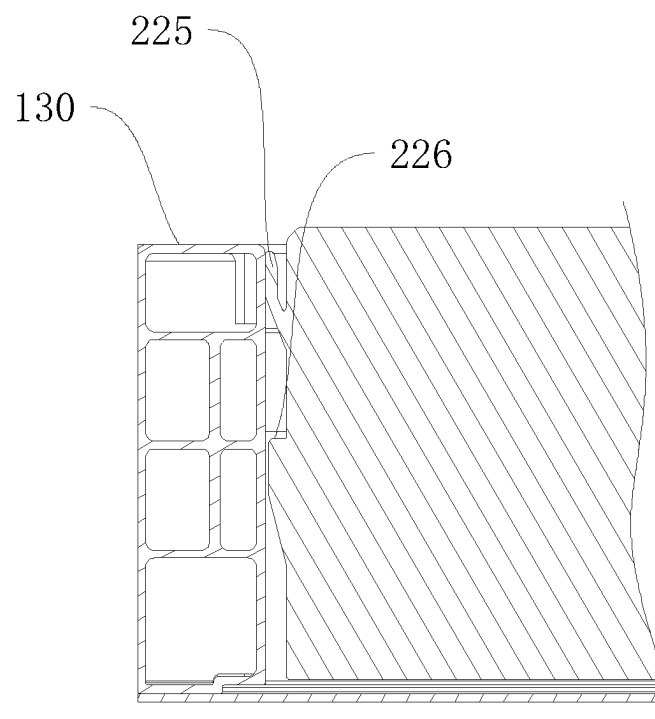


FIG. 14

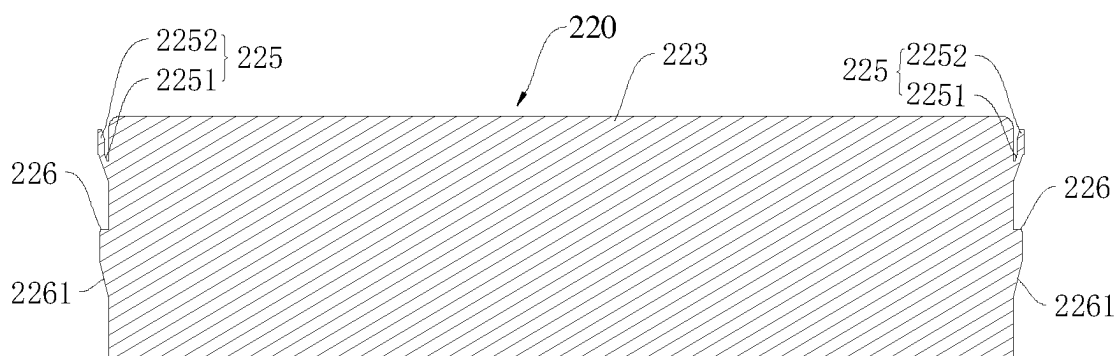


FIG. 15

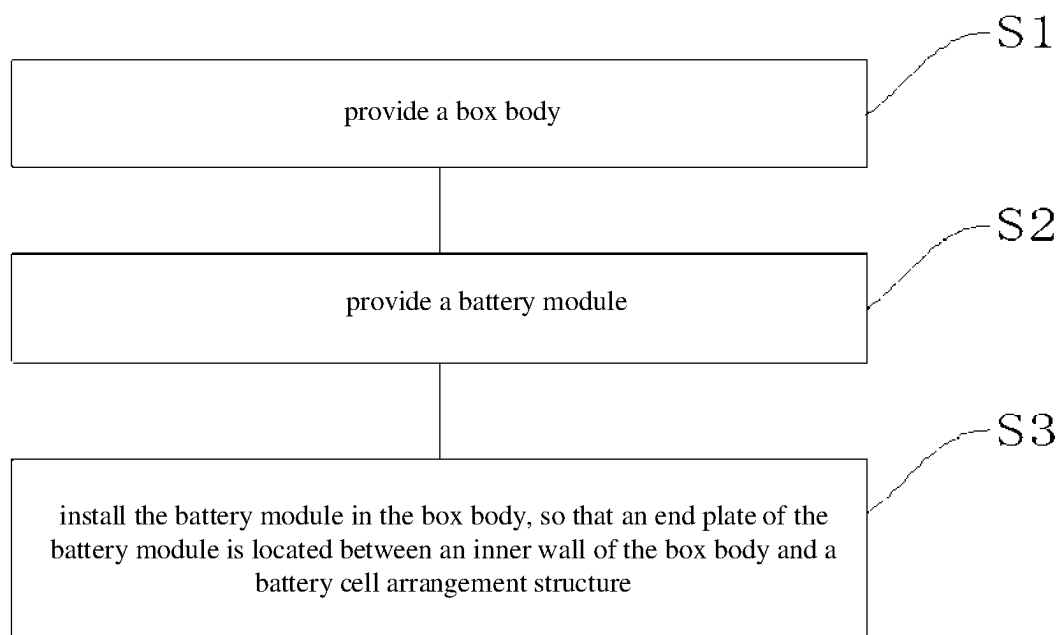


FIG. 16

1

BATTERY, POWER CONSUMPTION DEVICE, METHOD FOR PRODUCING BATTERY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Application No. PCT/CN2020/135942, filed on Dec. 11, 2020, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present application relates to a technical field of batteries, and in particular, to a battery, a power consumption device, and a method for producing a battery.

BACKGROUND

In the prior art, bolt locking is adopted between a battery module and a box body of a battery, so that a rigid connection is formed between the battery module and the box body. During expansion of the battery module, the box body is possibly deformed under an action of expansion force, which causes the battery box deforms of which a size of outline is increased, and affects assembly and a service life of the battery. In addition, a battery cell can be pressed by an inner wall of the box body during the expansion, which may cause occurrence of a lithium evolution phenomenon, resulting in a shrinkage of battery capacity.

SUMMARY

Embodiments of the present application aim to provide a battery, a power consumption device, and a method for producing a battery. In the battery, it can provide expansion space for the battery module, and a box body is not easily deformed.

In a first aspect, the embodiments of the present application provide a battery, including a box body and a battery module, the battery module is arranged in the box body, and the battery module includes a battery cell arrangement structure and an end plate, the battery cell arrangement structure includes a plurality of battery cells stacked along a first direction, the end plate is arranged between an inner wall of the box body and the battery cell arrangement structure, and the end plate is fixedly connected to the battery cell arrangement structure, where the end plate includes a first elastic support portion, and the first elastic support portion is configured to be capable of abutting against the inner wall of the box body and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands, to provide expansion space for the battery cell arrangement structure.

In the above technical solution, the first elastic support portion deforms itself to provide the expansion space and deformation is reliable. It can provide the expansion space for the battery cell arrangement structure in time, that is, provide the expansion space for the battery module, to release expansion force of the battery cell arrangement structure, and can reduce a possibility of the deformation of the box body caused by the expansion force, which improves installation and assembly reliability and a service life of the box body. Meanwhile, by releasing the expansion force, a possibility of occurrence of a lithium evolution of the battery cell due to excessive extrusion force between the inner wall

2

of the box body and the end plate can be reduced, which is conducive to a normal operation of the battery.

Optionally, the box body includes a first wall and a second wall, the second wall is connected to the first wall and extends upward, the battery module is located above the first wall, and the end plate is arranged between the second wall and the battery cell arrangement structure; and the first elastic support portion is configured to be capable of abutting against the second wall and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands.

In the above technical solution, the expansion space is provided between the end plate and the second wall of the box body for the battery cell arrangement structure, which can reduce the extrusion force of the battery module on the second wall of the box body.

Optionally, at least a part of the first elastic support portion extends obliquely upward towards the second wall.

In the above technical solution, when the end plate is pressed, since the first elastic support portion has an inclination angle, the first elastic support portion is more easily deformed when being pressed by the second wall, which can provide the expansion space for the battery cell arrangement structure in time.

Optionally, the end plate further includes an end plate body, the end plate body has a first surface facing the battery cell arrangement structure and a second surface facing away from the battery cell arrangement structure, and the first elastic support portion is arranged on the second surface.

In the above technical solution, the second surface is a large surface of the end plate body, which is convenient to arrange the first elastic support portion. In addition, the second surface is the large surface, which it is convenient to arrange more first elastic support portions, it is conducive to dispersing the expansion force of the battery cell arrangement structure, and reducing the possibility of the occurrence of the lithium evolution caused by compression on the battery cell due to the concentrated extrusion force.

Optionally, an orthographic projection of the first elastic support portion on a horizontal plane is a long-strip shape, a surface on which a long side of the long-strip shape is located is connected to the second surface, and a surface on which the other long side of the long-strip shape is located is configured to abut against the second wall, so as to disperse the expansion force of the battery cell arrangement structure in a length direction of the end plate.

Optionally, there are a plurality of the first elastic support portions, and the plurality of the first elastic support portions are arranged at intervals on the second surface along the up and down direction, so as to facilitate uniform dispersion of the expansion force and improve stress consistency of each place on a plane where the battery cell arrangement structure is in contact with the end plate, which improves the stress consistency of the battery cell.

Optionally, there are a plurality of the first elastic support portions, and the plurality of the first elastic support portions are arranged in a rectangular array on the second surface, so as to facilitate the uniform dispersion of the expansion force transmitted from the battery cell arrangement structure to the end plate, to further improve the stress consistency of each place of the battery cell arrangement structure, which improves the stress consistency of the battery cell.

Optionally, the box body includes a pair of second walls, the pair of second walls are arranged opposite to each other along the first direction, and the first elastic support portion abuts against the second wall when the battery cell arrange-

3

ment structure is not expanded, to implement a positioning of the battery module in the first direction.

In the above technical solution, when the battery module is installed in the box body, the first elastic support portion provides an installation positioning in the first direction for the battery module, which ensures installation reliability of the battery module in the first direction.

Optionally, the battery module is in interference fit with the box body, and the first elastic support portion is configured to absorb interference magnitude in the first direction by producing elastic deformation.

In the above technical solution, when the battery cell arrangement structure is not expanded, the first elastic support portion abuts against the second wall and is in a deformed state. In addition to providing the installation positioning in the first direction for the battery module as mentioned above, this arrangement also has at least the following two advantages: first, after the battery module is assembled in place, the first elastic support portion provides assembly allowance in the first direction for the battery module, so that a size error of the battery module in the first direction can be offset by the deformation of the first elastic support portion, for example, when a size of the battery module in the first direction is larger than an installation size of the box body in a corresponding direction, the battery module can be smoothly loaded in the box body through the deformation of the first elastic support portion, and thus requirements on processing and assembly accuracy of the battery module in the first direction are reduced; second, when the battery module is assembled in place, since the first elastic support portion is in the deformed state, reaction force of the second wall can be sent to the battery cell arrangement structure, as the battery cell is subjected to a certain amount of pressure, it is conducive to ensuring a good contact of interfaces of a positive electrode sheet and a negative electrode sheet inside the battery cell.

Optionally, the box body further includes a pair of third walls, the pair of third walls are connected to the first wall and extend upward, the pair of third walls are arranged opposite to each other along a second direction, and the second direction intersects the first direction; and the end plate further includes a second elastic support portion, and the second elastic support portion is used to abut against the third wall, to implement a positioning of the battery module in the second direction.

In the above technical solution, when the battery module is installed in the box body, the second elastic support portion abuts against the third wall, which provides an installation positioning in the second direction for the battery module, and ensures the installation reliability of the battery module in the second direction.

Optionally, the battery module is in interference fit with the box body, and the second elastic support portion is configured to absorb interference magnitude in the second direction by producing elastic deformation.

In the above technical solution, when the battery cell arrangement structure is not expanded, the second elastic support portion abuts against the third wall and is in the deformed state. In addition to providing the installation positioning in the second direction for the battery module as mentioned above, this arrangement also has at least the following two advantages: first, after the battery module is assembled in place, the second elastic support portion provides the assembly allowance in the second direction for the battery module, so that the size error of the battery module in the second direction can be offset by the deformation of the second elastic support portion, for example, when the

4

size of the battery module in the second direction is larger than the installation size of the box body in a corresponding direction, the battery module can be smoothly loaded in the box body through the deformation of the second elastic support portion, and thus the requirements on the processing and assembly accuracy of the battery module in the second direction are reduced; second, when the battery module is assembled in place, since the second elastic support portion is in the deformed state, the reaction force of the second wall can be sent to the battery cell arrangement structure, as the battery cell is subjected to the certain amount of the pressure, it is conducive to ensuring the good contact of the interfaces of the positive electrode sheet and the negative electrode sheet inside the battery cell.

Optionally, the end plate further comprises an end plate body, and the second elastic support portion is arranged on both opposite sides of the end plate body along the second direction.

Optionally, at least a part of the second elastic support portion extends upward from the end plate body towards the third wall.

In the above technical solution, an upward extending part of the second elastic support portion can abut against the third wall to implement the installation positioning of the second elastic support portion and the third wall, and the second elastic support portion extends obliquely upward, which is conducive to the second elastic support portion being pressed and deformed by the third wall.

Optionally, the second elastic support portion includes a first section and a second section, the first section extends obliquely upward from the end plate body towards the third wall, the second section extends upward from an end of the first section away from the end plate body, and the second section is used to abut against the third wall.

In the above technical solution, the first section is arranged obliquely upward, which can play a certain guide role. The second section extends along the up and down direction, and can form surface contact with the third wall, which is conducive to extrusion and fitting of the two. Moreover, the second section has a gap with the end plate body, which is conducive to the deformation of the second section when being pressed by the third wall.

Optionally, the end plate further includes a guide portion, the guide portion is arranged on both opposite sides of the end plate body along the second direction, the guide portion is located below the second elastic support portion, the guide portion has a guide chamfer, and the guide chamfer is used to guide when the end plate is loaded in the box body.

Optionally, the guide portion has a gap with the third wall, so as to prevent the guide portion from affecting the normal operation of the second elastic support portion.

Optionally, a lower end of the battery cell arrangement structure is attached to the first wall, and a lower end of the end plate has a gap with the first wall, so as to ensure contact between the battery cell arrangement structure and the first wall.

In a second aspect, the embodiments of the present application provide a power consumption device, the power consumption device includes the battery provided by the first aspect according to the embodiments of the present application.

In a third aspect, the embodiments of the present application provide a method for producing a battery, including: providing a box body; providing a battery module, the battery module includes a battery cell arrangement structure and an end plate, the battery cell arrangement structure comprises a plurality of battery cells stacked

5

upon each other, the end plate is fixedly connected to the battery cell arrangement structure, the end plate includes a first elastic support portion, the first elastic support portion is configured to be capable of abutting against an inner wall of the box body and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands, to provide expansion space for the battery cell arrangement structure; and installing the battery module in the box body, so that the end plate is located between the inner wall of the box body and the battery cell arrangement structure.

BRIEF DESCRIPTION OF DRAWINGS

In order to illustrate the technical solution in embodiments of the present application more clearly, brief description will be made below to the drawings required in the embodiments, and it should be understood that the following drawings only show some embodiments of the present application, therefore, should not to be regarded as the restriction to the scope, and other related drawings could be obtained based on these drawings by those ordinary skilled in this art without creative efforts.

FIG. 1 is a schematic diagram of a vehicle provided by an embodiment of the present application;

FIG. 2 is an exploded schematic diagram of a battery provided by an embodiment of the present application;

FIG. 3 is an exploded schematic diagram of a battery provided by an embodiment of the present application, where an upper cover body is not shown;

FIG. 4 is a left side schematic diagram of a battery provided by an embodiment of the present application;

FIG. 5 is a cross-sectional schematic diagram along A-A line in FIG. 4;

FIG. 6 is a partially enlarged schematic diagram of FIG. 5;

FIG. 7 is a cross-sectional schematic diagram along a first direction of an end plate provided by an embodiment of the present application;

FIG. 8 is a stereoscopic schematic structural diagram of an end plate provided by an embodiment of the present application;

FIG. 9 is a side view schematic diagram of an end plate provided by an embodiment of the present application;

FIG. 10 is a side view schematic diagram of an end plate provided by another embodiment of the present application;

FIG. 11 is a stereoscopic structural schematic diagram of a box body of a battery provided by an embodiment of the present application, where an upper cover body is not shown;

FIG. 12 is a main view schematic diagram of a battery provided by an embodiment of the present application;

FIG. 13 is a cross-sectional schematic diagram along B-B line in FIG. 12;

FIG. 14 is a partially enlarged schematic diagram of FIG. 13;

FIG. 15 is a cross-sectional schematic diagram along a second direction of an end plate provided by an embodiment of the present application;

FIG. 16 is a schematic flowchart of a method for producing a battery provided by an embodiment of the present application.

DESCRIPTION OF EMBODIMENTS

In order to make the objectives, technical solutions and advantages of embodiments of the present application

6

clearer, the technical solutions in the embodiments of the present application will be clearly and completely described below with reference to the accompanying drawings in the embodiments of the present application. Apparently, the described embodiments are some of, rather than all of, the embodiments of the present application. Components of the embodiments of the present application generally described and shown in the accompanying drawings herein may be arranged and designed in various different configurations.

Therefore, the following detailed description of the embodiments of the present application provided in the accompanying drawings is not intended to limit the scope of the present application claimed for protection, but merely indicates the selected embodiments of the present application. All the other embodiments obtained by those ordinary skilled in the art based on the embodiments of the present application without any creative effort shall fall within the scope of protection of the present application.

It should be noted that, the embodiments in the present application and features in the embodiments may be mutually combined provided that no conflict is caused. It should be noted that similar reference numerals and letters indicate similar items in the following accompany drawings. Therefore, once a certain item is defined in one drawing, it does not need to be further defined and explained in the subsequent drawings.

In the description of the embodiments of the present application, it should be noted that orientations or positional relationships indicated by terms such as “up”, “down”, “inside” and “outside” are orientations or positional relationships shown based on the accompany drawings, or the orientations or positional relationships commonly placed when the application product is used, or the orientations or positional relationships commonly understood by those skilled in the art, or the orientations or positional relationships commonly placed when the application product is used, and the terms are merely for convenience of describing the present application and for simplifying the description, but for indicating or implying that an indicated apparatus or element must have a specific orientation, and must be constructed and operated in the specific orientation, which thus may not be understood as limiting the present application. In addition, the terms “first”, “second”, “third” and the like are only intended for distinguishing description, and shall not be understood as an indication or implication of relative importance.

In the description of the present application, it also should be noted that unless otherwise explicitly specified and defined, the terms “setting”, “mounting”, “connecting” and “connection” should be understood in a broad sense, for example, they may be a fixed connection, a detachable connection, or an integrated connection, may be a mechanical connection, or may be an electrical connection; may be a direct connection and may also be an indirect connection via an intermediate medium, or may be communication between the interiors of two elements. Those ordinary skilled in the art can appreciate the specific meanings of the foregoing terms in the present application according to specific conditions.

An application of a battery generally includes three levels: a battery cell, a battery module and a battery pack. The battery cell includes a positive electrode plate, a negative electrode plate, an electrolyte and a separator. The separator is arranged between the positive electrode plate and the negative electrode plate to prevent an internal short circuit. Generally, the battery cell can be divided into three types

according to a packaging mode: a cylindrical battery cell, a square battery cell and a soft pack battery cell.

The battery module refers to a single physical module including a plurality of battery cells to provide a higher voltage and/or capacity. In the battery module, the plurality of battery cells can be connected in series and/or in parallel via a bus bar, so as to be used in various applications, for example, in some high-power applications such as electric vehicles.

The battery pack is constructed by assembling components such as a battery management system on the basis of one or more battery modules, and loading the components in a sealed box body, the sealed box body is then connected to power consumption devices such as the electric vehicles. The battery mentioned in the present application can be the battery pack.

In the battery of the prior art, the battery module and the box body are rigidly connected with a fastener, and during expansion of the battery module, a part of expansion force can be released to a certain extent through local deformation of the end plate. However, due to use of bolt locking between the end plate and the box body, a rigid connection is formed between the two. Therefore, the box body is bound to be affected during deformation of the end plate, which may cause the deformation of the box body. However, if it is desired to avoid the deformation of the box body due to the deformation of the end plate, the local deformation of the end plate can be small, which is not conducive to providing sufficient expansion space for the battery module. And the battery cell can still be pressed by an inner wall of the box body during the expansion, which may cause occurrence of a lithium evolution phenomenon, resulting in a shrinkage of battery capacity.

In view of this, some embodiments of the present application provide a battery 10. In the battery 10, expansion space can be provided for a battery module 200 to release expansion force, which is conducive to decreasing a possibility of the occurrence of a lithium evolution caused by compression on the battery cell 211, and also decreasing a possibility of deformation of a box body 100 at the same time. In other words, the possibility of the deformation of the box body 100 can be reduced while providing the expansion space for the battery module 200.

The embodiments of the present application provides a power consumption device using the battery 10 as a power supply, and the power consumption device can be, but is not limited to, a vehicle 1, a ship or an aircraft.

It can be understood that the battery 10 described in the embodiments of the present application is applicable to various devices using batteries, such as mobile phones, notebook computers, battery carts, electric vehicles, ships, spacecrafts, electric toys and electric tools, etc., for example, the spacecrafts include rockets, space shuttles and space-ships, etc.; the electric toys include fixed or mobile electric toys, such as game consoles, electric vehicle toys, electric ship toys and electric airplane toys, etc.; the electric tools include metal cutting power tools, grinding power tools, assembly power tools and railway power tools, such as electric drills, electric grinders, electric wrenches, electric screwdrivers, electric hammers, concrete vibrators, and electric planers.

The battery 10 described in the embodiments of the present application is not only applicable to the power consumption device described above, but also applicable to all devices that use the battery 10.

As shown in FIG. 1, FIG. 1 is a structural schematic diagram of a vehicle 1 according to an embodiment of the

present application. The vehicle 1 can be a fuel-powered vehicle, a gas-powered vehicle or a new energy vehicle, and the new energy vehicle can be a battery electric vehicle, a hybrid vehicle, an extended-range vehicle, or the like. The battery 10, a motor 20 and a controller 30 can be arranged in the vehicle 1, and the controller 30 is used to control the battery 10 to supply power to the motor 20, for example, the battery 10 is arranged at the bottom or the head of the vehicle 1. The battery 10 can be used to supply power to the vehicle 1, for example, the battery 10 can be used as an operation power supply of the vehicle 1 and is used for a circuit system of the vehicle 1, for example, for a working power demand of the vehicle 1 during startup, navigation and operation.

In another embodiment of the present application, the battery 10 can be used not only as the operation power supply of the vehicle 1 but also as a driving power supply of the vehicle 1, replacing or partially replacing fuel or natural gas to provide driving power for the vehicle 1.

In some embodiments of the present application, the vehicle can be powered by the battery 10 as shown in FIG. 2. The battery 10 includes the battery module 200 and the box body 100, and the battery module 200 is arranged in the box body 100. As shown in FIG. 2 and FIG. 3, the battery module 200 includes a battery cell arrangement structure 210 and an end plate 220, the battery cell arrangement structure 210 can include a plurality of battery cells 211 stacked along a first direction A1, the end plate 220 is arranged between an inner wall of the box body 100 and the battery cell arrangement structure 210, and the end plate 220 is fixedly connected to the battery cell arrangement structure 210.

Among them, as shown in FIGS. 4-6, the end plate 220 includes a first elastic support portion 224, and the first elastic support portion 224 is configured to be able to abut against the inner wall of the box body 100 and be pressed and deformed by the battery cell arrangement structure 210 when the battery cell arrangement structure 210 expands, so as to provide expansion space for the battery cell arrangement structure 210.

In the above technical solution, since the battery cell arrangement structure 210 includes a plurality of battery cells 211 in the first direction A1, the battery module 200 mainly expands along the first direction A1 when expanding. The first elastic support portion 224 deforms itself to provide the expansion space and the deformation is reliable, which can provide the expansion space for the battery cell arrangement structure 210 in time (that is, provide the expansion space for the battery module 200), mainly providing the expansion space in the first direction, so as to release expansion force of the battery cell arrangement structure 210, and be capable of reducing the possibility of the deformation of the box body 100 caused by the expansion force, which improves installation and assembly reliability and a service life of the box body 100. Meanwhile, by releasing the expansion force, the possibility of the occurrence of the lithium evolution of the battery cell 211 due to excessive extrusion force between the inner wall of the box body 100 and the end plate 220 can be reduced. Therefore, the possibility of the deformation of the box body 100 can be reduced while providing the expansion space for the battery module 200, which is conducive to a normal operation of the battery 10.

The end plate 220 and the battery cell arrangement structure 210 can be fixedly connected in any suitable manner, for example, bonding, strap connection, or using an end side plate to connect the end plate 220 and the battery

cell arrangement structure 210, and the embodiments of the present application are not limited thereto.

In an embodiment of the present application, as shown in FIG. 2 and FIG. 3, the box body 100 can include a lower box body 101 and an upper cover body 102, the upper cover body 102 is sealed and covered on the lower box body 101, and the battery module 200 can be installed in the lower box body 101.

As shown in FIGS. 3-6, in an embodiment of the present application, the box body 100 includes a first wall 110 and a second wall 120, the second wall 120 is connected to the first wall 110 and extends upward, the battery module 200 is located above the first wall 110, that is, the first wall 110 is a bottom wall of the box body 100, the second wall 120 is a side wall connected to the bottom wall, the first wall 110 can be a bottom wall of the lower box body 101, and the second wall 120 can be a side wall of the lower box body 101. The end plate 220 is arranged between the second wall 120 and the battery cell arrangement structure 210. The first elastic support portion 224 is configured to be able to abut against the second wall 120 and be pressed and deformed by the battery cell arrangement structure 210 when the battery cell arrangement structure 210 expands. On the basis of this, the expansion space is provided between the end plate 220 and the second wall 120 of the box body 100 for the battery cell arrangement structure 210, which can reduce the extrusion force of the battery module 200 on the second wall 120 of the box body 100.

In order that the first elastic support portion 224 can be easily deformed when being compressed, as shown in FIG. 6, in an embodiment of the present application, at least a part of the first elastic support portion 224 extends obliquely upward towards the second wall 120. On the basis of this, when the end plate 220 is compressed, the first elastic support portion 224 has an inclination angle, so that the first elastic support portion 224 is more easily deformed when being pressed by the second wall 120, which can provide the expansion space for the battery cell arrangement structure 210 in time.

It should be noted that an upward inclination angle of the first elastic support portion 224 can be any angle as long as it can meet deformation needs. The embodiments of the present application do not limit the upward inclination angle of the first elastic support portion 224.

As shown in FIG. 6 and FIG. 7, in an embodiment of the present application, the end plate 220 further includes an end plate body 223, the end plate body 223 has a first surface 2231 facing the battery cell arrangement structure 210 and a second surface 2232 facing away from the battery cell arrangement structure 210, and the first elastic support portion 224 is arranged on the second surface 2232. The second surface 2232 is a large surface of the end plate body 223, which is convenient to arrange the first elastic support portion 224. In addition, since the second surface 2232 is the large surface (the surface with the larger area), which is convenient to arrange more first elastic support portions 224, it is conducive to dispersing the expansion force of the battery cell arrangement structure 210, and reducing the possibility of the occurrence of the lithium evolution caused by the compression on the battery cell 211 due to the concentrated extrusion force.

It can be understood that in other embodiments of the present application, the first elastic support portion 224 can also be arranged on two side walls in a thickness direction (the first direction A1) of the end plate body 223, that is,

arranged on a small surface (the surface with a smaller area) of the end plate body 223, and extend towards the second wall 120.

In order to disperse the expansion force of the battery cell arrangement structure 210 as much as possible, in an embodiment of the present application, an orthographic projection of the first elastic support portion 224 on a horizontal plane is a long-strip shape, a surface on which a long side of the long-strip shape is located is connected to the second surface 2232, and a surface on which the other long side of the long-strip shape is located is used to abut against the second wall 120. In other words, the first elastic support portion 224 is arranged on the second surface 2232 along the length direction of the end plate 220 (that is, a second direction A2, which intersects the first direction A1). In this way, it is conducive to dispersing the expansion force of the battery cell arrangement structure 210 in the length direction of the end plate 220.

In order that the first elastic support portion 224 can be easily deformed, as shown in FIG. 6 and FIG. 7, optionally, the first elastic support portion 224 can be formed as a sheet shape. That is, in the embodiment, the first elastic support portion 224 is constructed as a sheet shape structure extending obliquely upward.

It should be noted that the embodiments of the present application do not limit a specific structure of the first elastic support portion 224. For example, the first elastic support portion 224 can also be constructed as a horizontal protrusion extending along the first direction A1, and the first elastic support portion 224 is provided with a groove or other strength weakening structures to weaken strength of the first elastic support portion 224, so that the first elastic support portion 224 is easily deformed when being pressed by the second wall 120. In addition, the first elastic support portion 224 can also be a spring, one end of the spring is connected to (for example, welded) the second surface 2232, and the other end extends towards the second wall 120.

As shown in FIGS. 6-9, in an embodiment of the present application, there are a plurality of the first elastic support portions 224, and the plurality of the first elastic support portions 224 are arranged at intervals on the second surface 2232 along the up and down direction, so as to facilitate uniform dispersion of the expansion force, and improve stress consistency of each place on a plane where the battery cell arrangement structure 210 is in contact with the end plate 220, which improves the stress consistency of the battery cell 211.

Further, as shown in FIGS. 6-9, the plurality of the first elastic support portions 224 can be arranged in a rectangular array on the second surface 2232, which is conducive to dispersing the expansion force transmitted from the battery cell arrangement structure 210 to the end plate 220, to further improve the stress consistency of each place of the battery cell arrangement structure 210 and improve the stress consistency of the battery cell 211.

As shown in FIG. 10, in another embodiment of the present application, there are the plurality of the first elastic support portions 224, the plurality of the first elastic support portions 224 are arranged at intervals on the second surface 2232 along the up and down direction, each first elastic support portion 224 extends in the length direction of the end plate body 223, and the extension of the first elastic support portions 224 in the length direction of the end plate body 223 is not interrupted.

In the battery of the prior art, in order to install the battery module 200 in the box body 100 and restrict the battery module 200 from freely moving upward in the box body

11

100, the end plate 220 of the battery module 200 is generally installed on a side wall of the box body 100 through a fastener (such as a locking bolt). In order that the fastener can be easily installed, a thickness size of the side wall of the box body is relatively large. Therefore, it is not conducive to increasing space of the box body 100 for accommodating the battery module 200, which reduces space utilization in the box body 100.

In view of this, as shown in FIG. 6, in an embodiment of the present application, a restraint surface 121 is arranged on the second wall 120, the restraint surface 121 is used to abut against the end plate 220 to restrict the end plate 220 from moving upward. The restraint surface 121 restricts the end plate 220 from freely moving upward, which ensures installation locations of the end plate 220 and the battery module 200 in the up and down direction (that is, a height direction of the end plate 220), and avoids affecting the normal operation of the battery module 200 due to an upward movement of the battery module 200. On the basis of this, the bolt locking between the battery module 200 and the box body 100 can be omitted, which is conducive to reducing use of parts and increasing the space utilization in the box body 100. Moreover, the bolt locking is canceled, a rigid connection between the battery module 200 and the box body 100 is released, so the possibility of the deformation of the box body 100 can be reduced while providing the expansion space for the battery module 200, and at the same time, canceling the bolt locking simplifies the process, which improves assembly efficiency of the battery 10.

The restraint surface 121 can be constructed by any suitable structure. As shown in FIG. 6, in an embodiment of the present application, the second wall 120 is partially recessed towards a surface of the end plate 220 to form a groove, and an upper side wall of the groove is the restraint surface 121. The upper side wall of the groove constructs the restraint surface 121, which not only does not occupy the space of the box body 100 for accommodating the battery module 200, but also is conducive to lightweight of the box body 100.

As shown in FIG. 6, in an embodiment of the present application, the end plate 220 is provided with a restraint protrusion 221, and an upper surface of the restraint protrusion 221 is used to abut against the restraint surface 121 to restrict the end plate 220 from moving upward. Compared with the bolt locking, a manner that the restraint protrusion 221 abuts against and restrain the restraint surface 121 has advantages of a simple structure and a convenient installation of the battery module 200 in the box body 100 while implementing reliable restraint between the end plate 220 and the box body 100.

In order to ensure a restraint effect of the restraint protrusion 221, as shown in FIGS. 4-6, in an embodiment of the present application, the restraint protrusion 221 extends horizontally towards the second wall 120. On the basis of this, when the battery module 200 expands, a direction (upward) in which the restraint protrusion 221 may move is perpendicular to an extending direction, so that it is not easy to slide between the restraint protrusion 221 and the restraint surface 121, which can improve restraint reliability of the restraint protrusion 221 and the restraint surface 121.

Further, as shown in FIG. 6, the restraint surface 121 can be a surface parallel to the horizontal plane. In this way, when the restraint protrusion 221 abuts against the restraint surface 121, the two surfaces are fully fitted, which further improves the restraint reliability of the restraint protrusion 221 and the restraint surface 121.

12

When assembling the battery 10, first, the extrusion force can be applied to the battery module 200 to compress a length of the battery module 200 in the first direction A1; then the battery module 200 in a compressed state is placed in the box body 100 and disposed on the first wall 110; next, the extrusion force is removed to restore the length of the battery module 200, so that the restraint protrusion 221 of the end plate 220 moves to a lower part of the restraint surface 121.

In order that the restraint protrusion 221 can smoothly move to the lower part of the restraint surface 121 after the compression is removed, as shown in FIG. 6, in an embodiment of the present application, after the battery module 200 is installed in place in the box body 100, the upper surface of the restraint protrusion 221 has a gap with the restraint surface 121 in the up and down direction when the battery cell arrangement structure 210 is not expanded. On the basis of this, after the extrusion force pressing the end plate 220 is removed, since there is a gap in the up and down direction, the restraint surface 121 will not interfere with a horizontal movement of the restraint protrusion 221, which is conducive to the smooth movement of the restraint protrusion 221 to the lower part of the restraint surface 121.

As shown in FIG. 6, optionally, after the battery module 200 is installed in place in the box body 100, the restraint protrusion 221 also has a gap with the second wall 120 in a horizontal direction when the battery cell arrangement structure 210 is not expanded, so that the end plate 220 at least can move towards the second wall 120, so as to provide the expansion space for the battery cell arrangement structure 210.

In this way, during the expansion of the battery cell arrangement structure 210, the movement of the restraint protrusion 221 towards the second wall 120 can be divided into two stages. In a first stage, the restraint protrusion 221 moves horizontally towards the second wall 120 until it abuts on the second wall 120. If the battery cell arrangement structure 210 continues to expand after the restraint protrusion 221 abuts on the second wall 120, then the end plate body 223 can be locally deformed, so as to continue to provide the expansion space.

As shown in FIG. 6, the restraint protrusion 221 can be arranged on the second surface 2232 of the end plate body 223. The second surface 2232 is the large surface (the surface with the larger area) of the end plate body 223, which is convenient to arrange the restraint protrusion 221. Moreover, the second surface 2232 is the large surface, so it is convenient to arrange a longer restraint protrusion 221 along a length direction (a direction perpendicular to the first direction A1) of the end plate 220, so as to increase the restraint reliability of the battery module 200 in the box body 100 in the up and down direction as much as possible.

It can be understood that in other embodiments of the present application, the restraint protrusion 221 can also be arranged on two side surfaces in the thickness direction (the first direction A1) of the end plate body 223, that is, arranged on the small surface (the surface with the smaller area) of the end plate body 223, and extend towards the second wall 120.

In an embodiment of the present application, an orthographic projection of the restraint protrusion 221 on a level is a long-strip shape, a surface on which a long side of the long-strip shape is located is connected to the second surface 2232. The restraint protrusion 221 is the long-strip shape, which is conducive to increasing a contact area between the restraint protrusion 221 and the first restraint surface 121, so that it is conducive to enhancing the restraint effect of the restraint protrusion 221.

13

In an embodiment of the present application, the battery module **200** is bonded to the first wall **110** via glue. As shown in FIGS. **8-10**, a lower end of the end plate **220** is also provided with an extension portion **227**, which can play a role of blocking structural glue, and avoid the upward movement of the structural glue to a certain extent.

As shown in FIGS. **8-10**, in an embodiment of the present application, an upper end of the end plate **220** is also provided with a clamping portion **222**, which is used to facilitate an assembly tool to clamp the battery module **200** when the battery module **200** is loaded in the box body **100**.

As shown in FIG. **3** and FIG. **11**, in an embodiment of the present application, the box body **100** can include a pair of second walls **120**, the pair of second walls **120** are arranged opposite to each other along the first direction **A1**. The first elastic support portion **224** can abut against the second wall **120** when the battery cell arrangement structure **210** is not expanded, to implement a positioning of the battery module **200** in the first direction **A1**. On the basis of this, after the battery module **200** is installed in the box body **100**, the first elastic support portion **224** provides an installation positioning in the first direction **A1** for the battery module **200**, which ensures the installation reliability of the battery module **200** in the first direction **A1**.

It should be noted that in the embodiments of the present application, one end of the battery module **200** can abut against one of the second walls **120** via the first elastic support portion **224**, and the other end is fixedly connected to the other second wall **120**, or both ends of the battery module **200** abut against the corresponding second walls **120** via the first elastic support portion **224**.

Here, when the battery cell arrangement structure **210** is not expanded, the first elastic support portion **224** abuts against the second wall **120**, which can mean that the first elastic support portion **224** is just in contact with the second wall **120** and the first elastic support portion **224** is in a undeformed state, or can mean that the first elastic support portion **224** abuts against the second wall **120** and is in a deformed state, and the embodiments of the present application are not limited thereto.

Optionally, in an embodiment of the present application, the battery module **200** is in interference fit with the box body **100**, and the first elastic support portion **224** is configured to absorb interference magnitude in the first direction **A1** by producing elastic deformation. That is, in the embodiment, when the battery cell arrangement structure **210** is not expanded, the first elastic support portion **224** abuts against the second wall **120** and is in the deformed state. In addition to providing the installation positioning in the first direction **A1** for the battery module **200** as mentioned above, this arrangement also has at least the following two advantages: first, after the battery module **200** is assembled in place, the first elastic support portion **224** provides assembly allowance in the first direction **A1** for the battery module **200**, so that a size error of the battery module **200** in the first direction **A1** can be offset by the deformation of the first elastic support portion **224**, for example, when a size of the battery module **200** in the first direction **A1** is larger than an installation size of the box body **100** in a corresponding direction, the battery module **200** can be smoothly loaded in the box body **100** through the deformation of the first elastic support portion **224**, and thus requirements on processing and assembly accuracy of the battery module **200** in the first direction **A1** are reduced; second, when the battery module **200** is assembled in place, since the first elastic support portion **224** is in the deformed state, reaction force of the second wall **120** can be sent to the battery cell arrangement

14

structure **210**, as the battery cell **211** is subjected to a certain amount of pressure, it is conducive to ensuring a good contact of interfaces of a positive electrode sheet and a negative electrode sheet inside the battery cell **211**.

As shown in FIG. **11**, in an embodiment of the present application, the box body **100** further includes a pair of third walls **130**, the pair of third walls **130** are connected to the first wall **110** and extend upward, the pair of third walls **130** are arranged opposite to each other along a second direction **A2**, and the second direction **A2** intersects the first direction **A1**. Optionally, as shown in FIG. **11**, the second direction **A2** can be perpendicular to the first direction **A1**, and the second direction **A2** can be the length direction of the end plate **220**.

As shown in FIGS. **12-14**, the end plate **220** further includes a second elastic support portion **225**, and the second elastic support portion **225** is used to abut against the third wall **130**, to implement a positioning of the battery module **200** in the second direction **A2**. On the basis of this, when the battery module **200** is installed in the box body **100**, the second elastic support portion **225** abuts against the third wall **130** to provide an installation positioning in the second direction **A2** for the battery module **200**, which ensures the installation reliability of the battery module **200** in the second direction **A2**.

Here, when the battery cell arrangement structure **210** is not expanded, the second elastic support portion **225** abuts against the third wall **130**, which can mean that the second elastic support portion **225** is just in the contact with the third wall **130** and the second elastic support portion **225** is in the undeformed state, or can mean that the second elastic support portion **225** abuts against the third wall **130** and is in the deformed state, and the embodiments of the present application are not limited thereto.

Optionally, in an embodiment of the present application, the battery module **200** is in interference fit with the box body **100**, and the second elastic support portion **224** is configured to absorb interference magnitude in the second direction **A2** by producing elastic deformation. That is, in the embodiment, when the battery cell arrangement structure **210** is not expanded, the second elastic support portion **225** abuts against the second wall **130** and is in the deformed state. In addition to providing the installation positioning in the second direction **A2** for the battery module **200** as mentioned above, this arrangement also has at least the following two advantages: first, after the battery module **200** is assembled in place, the second elastic support portion **225** provides the assembly allowance in the second direction **A2** to the battery module **200**, so that the size error of the battery module **200** in the second direction **A2** can be offset by the deformation of the second elastic support portion **225**, for example, when the size of the battery module **200** in the second direction **A2** is larger than the installation size in of the box body **100** in a corresponding direction, the battery module **200** can be smoothly loaded in the box body **100** through the deformation of the second elastic support portion **225**, and thus the requirements on the processing and assembly accuracy of the battery module **200** in the second direction **A2** are reduced; second, when the battery module **200** is assembled in place, since the second elastic support portion **225** is in the deformed state, the reaction force of the third wall **130** can be sent to the battery cell arrangement structure **210**, as the battery cell **211** is subjected to the certain amount of the pressure, it is conducive to ensuring the good contact of the interfaces of the positive electrode sheet and the negative electrode sheet inside the battery cell **211**.

15

It should be noted that in the embodiments of the present application, the second elastic support portion 225 can be arranged between the battery module 200 and one of the third walls 130, that is, the second elastic support portion 225 is only arranged on one side of the end plate 220, or the second elastic support portion 225 can be arranged between the battery module 200 and the pair of third walls 130, that is, the second elastic support portion 225 is arranged on both opposite sides of the end plate 220.

Optionally, as shown in FIG. 13, in an embodiment of the present application, the second elastic support portion 225 is arranged on both opposite sides of the end plate body 223 in the second direction A2, that is, the end plate body 223 is provided with the second elastic support portion 225 on the both opposite sides along the second direction A2.

In the embodiments shown in FIGS. 11-14, a projection of an accommodating cavity of the box body 100 for accommodating the battery module 200 on the horizontal plane is a rectangle, the first direction A1 is a length direction of the rectangle, and the second direction A2 is a width direction of the rectangle. The battery module 200 is provided with the first elastic support portion 224 and the second elastic support portion 225 in the first direction A1 and the second direction A2, respectively, which can provide the reliable installation positioning for the battery module 200.

It can be understood that in other embodiments of the present application, the projection of the accommodating cavity of the box body 100 for accommodating the battery module 200 on the horizontal plane can be other shapes, such as a combination of the rectangle and a trapezoid.

As shown in FIG. 14 and FIG. 15, in an embodiment of the present application, at least a part of the second elastic support portion 225 extends upward from the end plate body 223 towards the second wall 130. In this way, an upward extending part of the second elastic support portion 225 can abut against the third wall 130 to implement the installation positioning of the second elastic support portion 225 and the third wall 130. In addition, the second elastic support portion 225 extends obliquely upward, which is conducive to the second elastic support portion 225 being pressed and deformed by the third wall 130.

The embodiments of the present application do not limit the specific structure of the second elastic support portion 225. Optionally, as shown in FIG. 15, in an embodiment of the present application, the second elastic support portion 225 includes a first section 2251 and a second section 2252, the first section 2251 extends obliquely upward from the end plate body 223 towards the third wall 130, the second section 2252 extends upward from an end of the first section 2251 away from the end plate body 223, and the second section is used to abut against the third wall 130. On the basis of this, the first section 2251 is arranged obliquely upward, which can play a certain guide role. The second section 2252 extends along the up and down direction, and can form surface contact with the third wall 130, which is conducive to extrusion and fitting of the two. Moreover, the second section 2252 has a gap with the end plate body 223, which is conducive to the deformation of the second section 2252 when being pressed by the third wall 130.

In other embodiments of the present application, the second elastic support portion 225 can be constructed as an elastic long strip extending along the up and down direction.

In order that the battery module 200 is smoothly loaded in the box body 100, as shown in FIG. 15, in an embodiment of the present application, the end plate 220 further includes a guide portion 226, the guide portion 226 is located below the second elastic support portion 225, the guide portion 226

16

has a guide chamfer 2261, and the guide chamfer 2261 is used to guide when the end plate 220 is loaded in the box body 100.

In order to prevent the guide portion 226 from affecting the normal operation of the second elastic support portion 225, as shown in FIG. 14, in an embodiment of the present application, the guide portion can have a gap with the third wall 130. In other words, as shown in FIG. 15, a height of the guide portion protruding from the end plate body 223 in the second direction A2 is smaller than a height of the second elastic support portion 225 protruding from the end plate body 223 in the second direction A2.

In the embodiments of the present application, a lower end of the battery cell arrangement structure 210 can be attached to the first wall 110, and a lower end of the end plate 220 has a gap with the first wall 110, so as to ensure contact between the battery cell arrangement structure 210 and the first wall 110.

It should be noted that "attach" in the above can mean that the lower end of the battery cell arrangement structure 210 is in contact with but not connected to the first wall 110, or can mean that the lower end of the battery cell arrangement structure 210 is in contact with and connected to the first wall 110, for example, the two are bonded via the glue.

As shown in FIG. 16, according to another aspect of the present application, a method for producing a battery 10 is provided, for example, the method for producing the battery 10 mentioned above, and the method includes the following steps:

S1: provide a box body 100;

S2: provide a battery module 120, the battery module 200 includes a battery cell arrangement structure 210 and an end plate 220, the battery cell arrangement structure 210 includes a plurality of battery cells 211 stacked upon each other, for example, includes a plurality of battery cells 211 stacked upon each other along a first direction A1, and the end plate 220 is fixedly connected to the battery cell arrangement structure 210; the end plate 220 includes a first elastic support portion 224, and the first elastic support portion 224 is configured to be capable of abutting against an inner wall of the box body 100 and be pressed and deformed by the battery cell arrangement structure 210 when the battery cell arrangement structure 210 expands, to provide expansion space for the battery cell arrangement structure 210;

S3: install the battery module 200 in the box body 100, so that the end plate 220 is located between the inner wall of the box body 100 and the battery cell arrangement structure 210.

Among them, the battery module 200 and the box body 100 can be the battery module 200 and the box body 100 mentioned above.

It should be noted that, the features in the embodiments of the present application may be mutually combined provided that no conflict is caused.

The above descriptions are only preferred embodiments of the present application, and are not intended to limit the present application. For those skilled in the art, the present application can have various modifications and changes. Any modification, equivalent replacement, improvement, or the like made within the spirit and principle of this application shall fall within the protection scope of this application.

What is claimed is:

1. A battery, comprising:
a box body;

17

a battery module, being arranged in the box body, comprising a battery cell arrangement structure and an end plate, the battery cell arrangement structure comprises a plurality of battery cells stacked along a first direction, the end plate is arranged between an inner wall of the box body and the battery cell arrangement structure, and the end plate is fixedly connected to the battery cell arrangement structure; and

wherein the end plate comprises a first elastic support portion, and the first elastic support portion is configured to be capable of abutting against the inner wall of the box body and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands, to provide expansion space for the battery cell arrangement structure; and

wherein the box body comprises a first wall and a second wall, the second wall is connected to the first wall and extends upward, the battery module is located above the first wall, and the end plate is arranged between the second wall and the battery cell arrangement structure; and

the first elastic support portion is configured to be capable of abutting against the second wall and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands; wherein the end plate further comprises an end plate body, the end plate body has a first surface facing the battery cell arrangement structure and a second surface facing away from the battery cell arrangement structure, and the first elastic support portion is arranged on the second surface; and wherein an orthographic projection of the first elastic support portion on a horizontal plane is a rectangle, a surface on which a long side of the rectangle is located is connected to the second surface, and a surface on which the other long side of the rectangle is located is configured to abut against the second wall.

2. The battery according to claim 1, wherein at least a part of the first elastic support portion extends obliquely upward towards the second wall.

3. The battery according to claim 1, wherein there are a plurality of the first elastic support portions, and the plurality of the first elastic support portions are arranged at intervals on the second surface along an up and down direction.

4. The battery according to claim 1, wherein there are a plurality of the first elastic support portions, and the plurality of the first elastic support portions are arranged in a rectangular array on the second surface.

5. The battery according to claim 1, wherein the box body comprises a pair of second walls, the pair of second walls are arranged opposite to each other along the first direction, and the first elastic support portion abuts against the second wall when the battery cell arrangement structure is not expanded, to implement a positioning of the battery module in the first direction.

6. The battery according to claim 1, wherein the box body further comprises a pair of third walls, the pair of third walls are connected to the first wall and extend upward, the pair of third walls are arranged opposite to each other along a second direction, and the second direction intersects the first direction; and

the end plate further comprises a second elastic support portion, and the second elastic support portion is configured to abut against the third wall, in order to implement a positioning of the battery module in the second direction.

18

7. The battery according to claim 6, wherein the battery module is in interference fit with the box body, and the second elastic support portion is configured to absorb interference magnitude in the second direction by producing elastic deformation.

8. The battery according to claim 6, wherein the end plate further comprises an end plate body, and the second elastic support portion is arranged on both opposite sides of the end plate body along the second direction.

9. The battery according to claim 8, wherein at least a part of the second elastic support portion extends upward from the end plate body towards the third wall.

10. The battery according to claim 8, wherein the second elastic support portion comprises a first section and a second section, the first section extends obliquely upward from the end plate body towards the third wall, the second section extends upward from an end of the first section away from the end plate body, and the second section is configured to abut against the third wall.

11. The battery according to claim 8, wherein the end plate further comprises a guide portion, the guide portion is arranged on both opposite sides of the end plate body along the second direction, the guide portion is located below the second elastic support portion, the guide portion has a guide chamfer, and the guide chamfer is configured to guide when the end plate is loaded in the box body.

12. The battery according to claim 11, wherein the guide portion has a gap with the third wall.

13. The battery according to claim 1, wherein a lower end of the battery cell arrangement structure is attached to the first wall, and a lower end of the end plate has a gap with the first wall.

14. The battery according to claim 1, wherein a restraint surface is arranged on the second wall, the restraint surface is used to abut against the end plate to restrict the end plate from moving upward, wherein

the restraint surface is an upper side wall of a groove on the second wall, and the groove is recessed towards a surface of the end plate,

or

the restraint surface is an upper surface of a restraint protrusion on the end plate.

15. The battery according to claim 1, wherein the battery module is in interference fit with the box body, and the first elastic support portion is configured to absorb interference magnitude in the first direction by producing elastic deformation.

16. A power consumption device, comprising: the battery according to claim 1.

17. A method for producing a battery, comprising:

providing a box body;

providing a battery module, the battery module comprises a battery cell arrangement structure and an end plate, the battery cell arrangement structure comprises a plurality of battery cells stacked upon each other, the end plate is fixedly connected to the battery cell arrangement structure, the end plate comprises a first elastic support portion, the first elastic support portion is configured to be capable of abutting against an inner wall of the box body; and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands, to provide expansion space for the battery cell arrangement structure;

installing the battery module in the box body, so that the end plate is located between the inner wall of the box body and the battery cell arrangement structure; and

19

wherein the box body comprises a first wall and a second wall, the second wall is connected to the first wall and extends upward, the battery module is located above the first wall, and the end plate is arranged between the second wall and the battery cell arrangement structure; 5
and
the first elastic support portion is configured to be capable of abutting against the second wall and be pressed and deformed by the battery cell arrangement structure when the battery cell arrangement structure expands; 10
wherein the end plate further comprises an end plate body, the end plate body has a first surface facing the battery cell arrangement structure and a second surface facing away from the battery cell arrangement structure, and the first elastic support portion is arranged on the 15
second surface;
and wherein an orthographic projection of the first elastic support portion on a horizontal plane is a rectangle, a surface on which a long side of the rectangle is located is connected to the second surface, and a surface on 20
which the other long side of the rectangle is located is configured to abut against the second wall.

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